

Recognizing the Use of Steganography in Forensic Evidence (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 02

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Time on Task:

5 hours, 45 minutes

Progress:

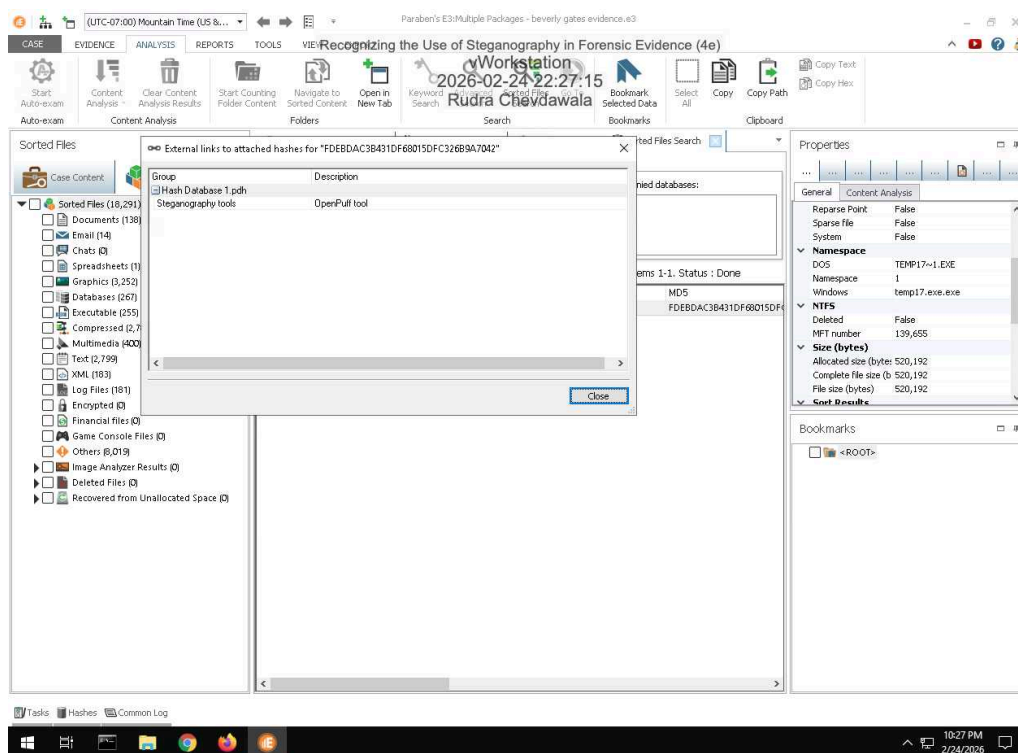
100%

Report Generated: Wednesday, February 25, 2026 at 3:31 PM

Section 1: Hands-On Demonstration

Part 1: Detect Steganography Software on a Drive Image

14. Make a screen capture showing the search result and its description.

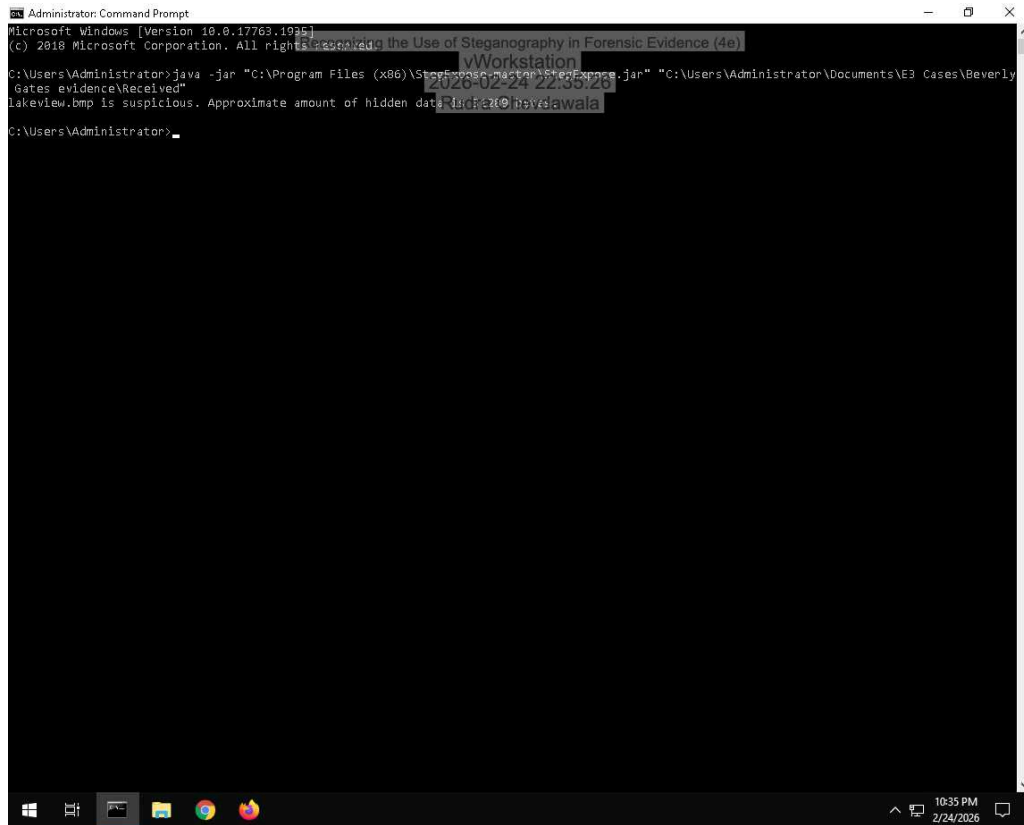


Part 2: Detect Hidden Data in Image Files

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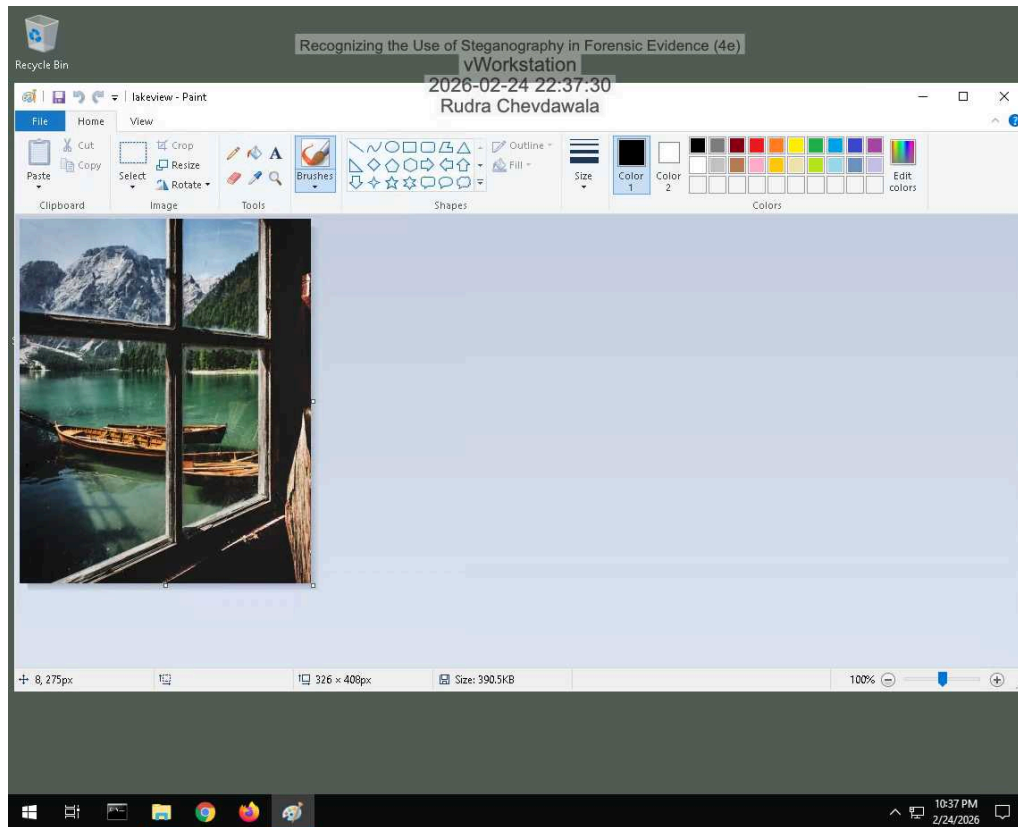
10. Make a screen capture showing the **StegExpose** results.



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13. Make a screen capture showing the suspicious file in Microsoft Paint.



Part 3: Extract Hidden Data from Image Files

2. Record the passphrase saved in the ReadMe file.

landmarks

16. Make a screen capture showing the contents of the file extracted by OpenPuff.



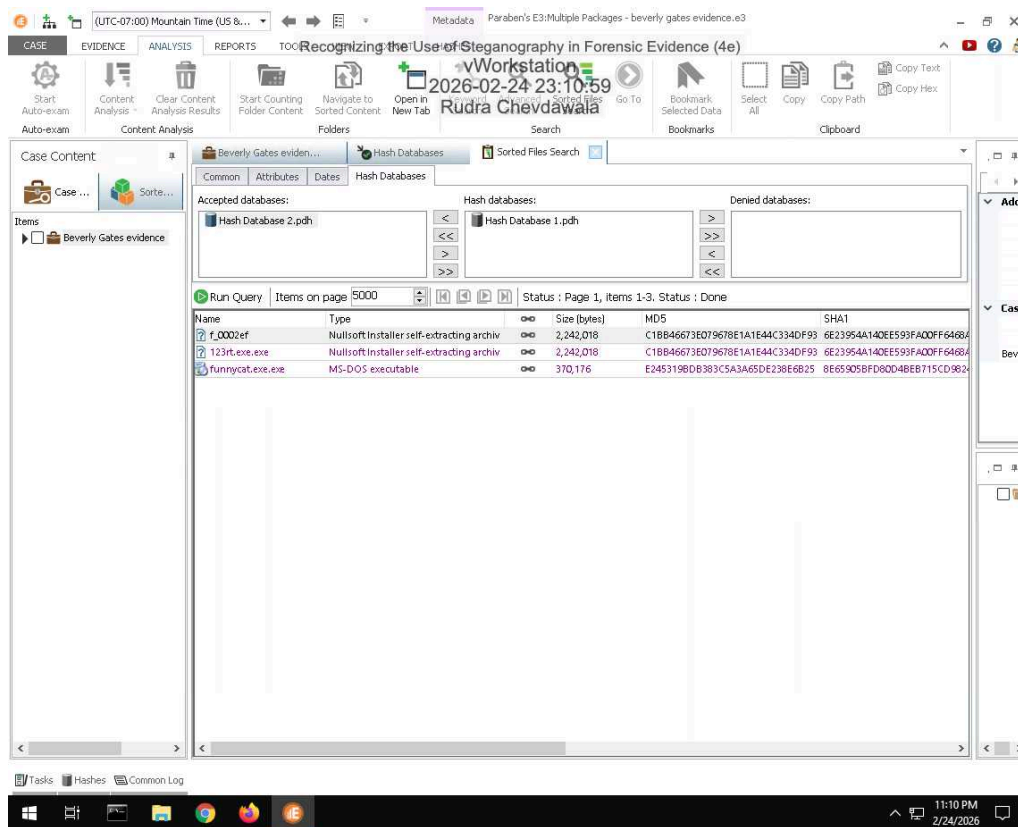
17. Describe the contents of the hidden file. How might it be relevant to the current investigation?

The file contains longitude and latitude coordinates for the city of Chicago. It might be relevant to the current investigation to show the exact locations, addresses, and evidence of the people involved in the case.

Section 2: Applied Learning

Part 1: Detect Steganography Software on a Drive Image

5. Make a screen capture showing the search result and its description.

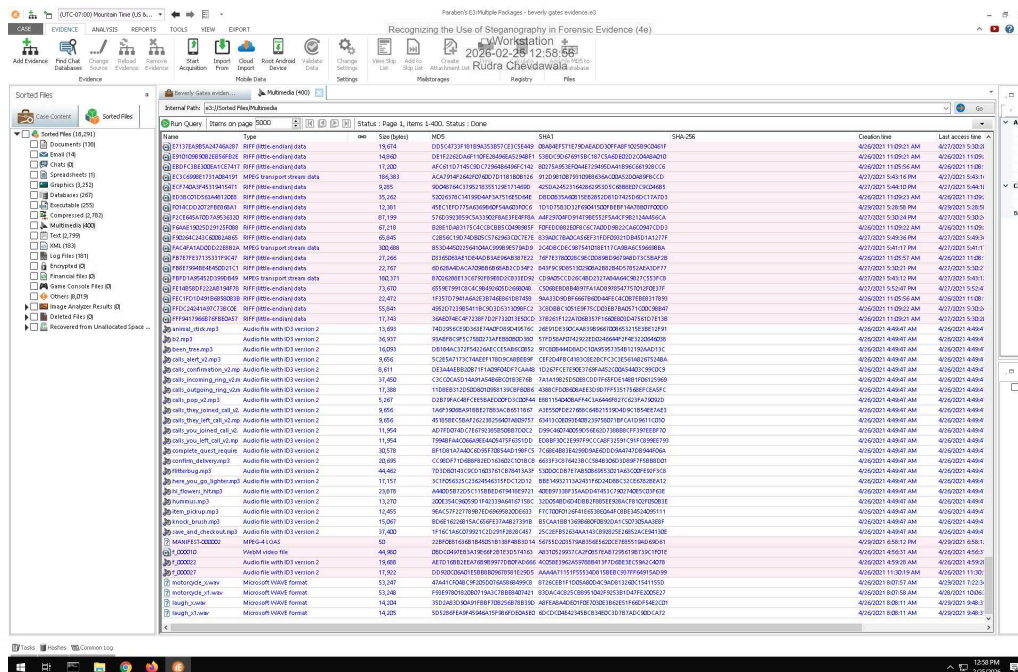


Part 2: Detect Hidden Data in Image and Audio Files

4. **Identify** the image file with concealed data according to the StegExpose steganalysis tool.

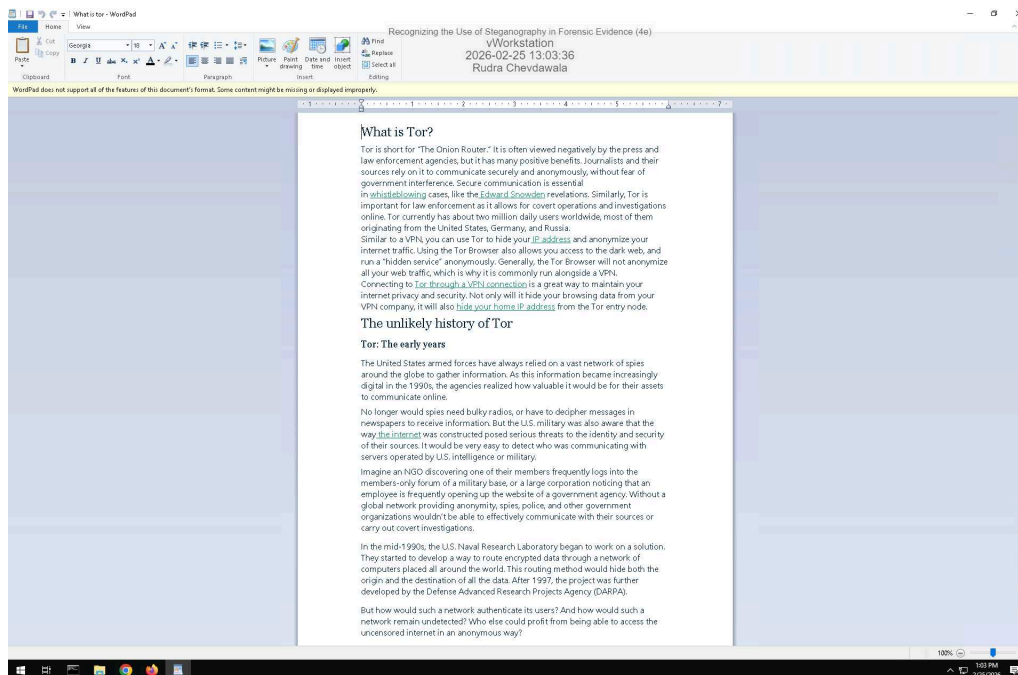
When executing the StegExpose Tool into CMD I get dB9solder.gif is suspicious. The file itself shows a small kitten with big eyes.

7. Make a screen capture showing the WAV file sizes and hash values in E3.

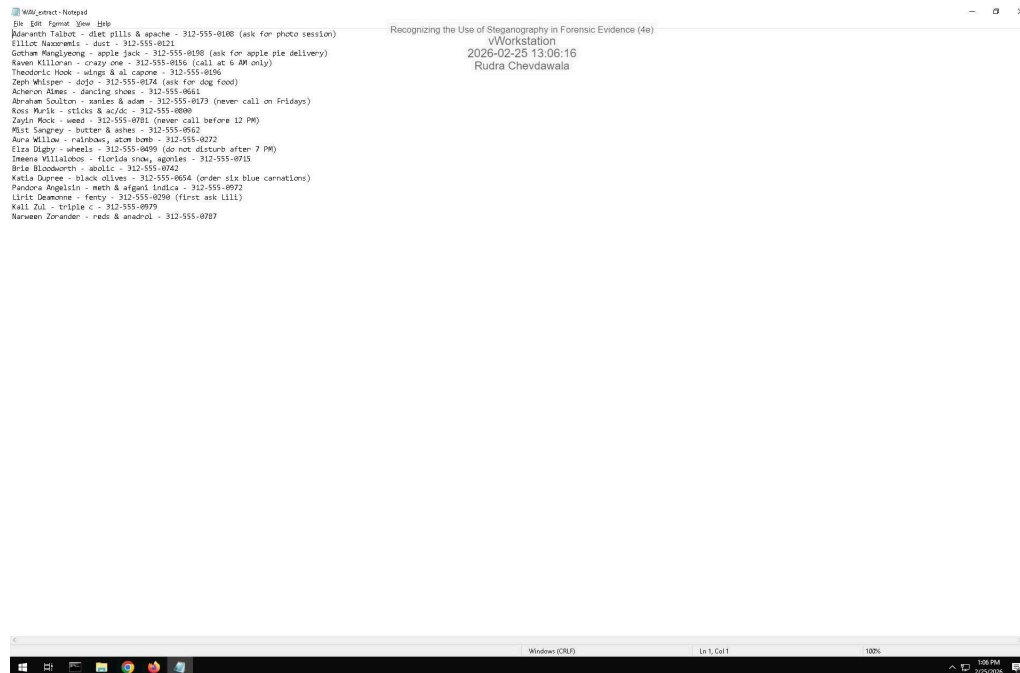


Part 3: Extract Hidden Data from Image and Audio Files

9. Make a screen capture showing the contents of the hidden file extracted by S-Tools.



15. Make a screen capture showing the contents of the hidden file extracted by Xiao.



16. Describe the contents of the two hidden files. How might they be relevant to the current investigation?

The first hidden file talks about Tor, which is usually used for buying things you can't buy on the regular internet through e-commerce on the dark web. The second hidden file lists the personal phone numbers of these fully named individuals, along with some brief descriptions.

Section 3: Challenge and Analysis

Part 1: Detect More Hidden Data

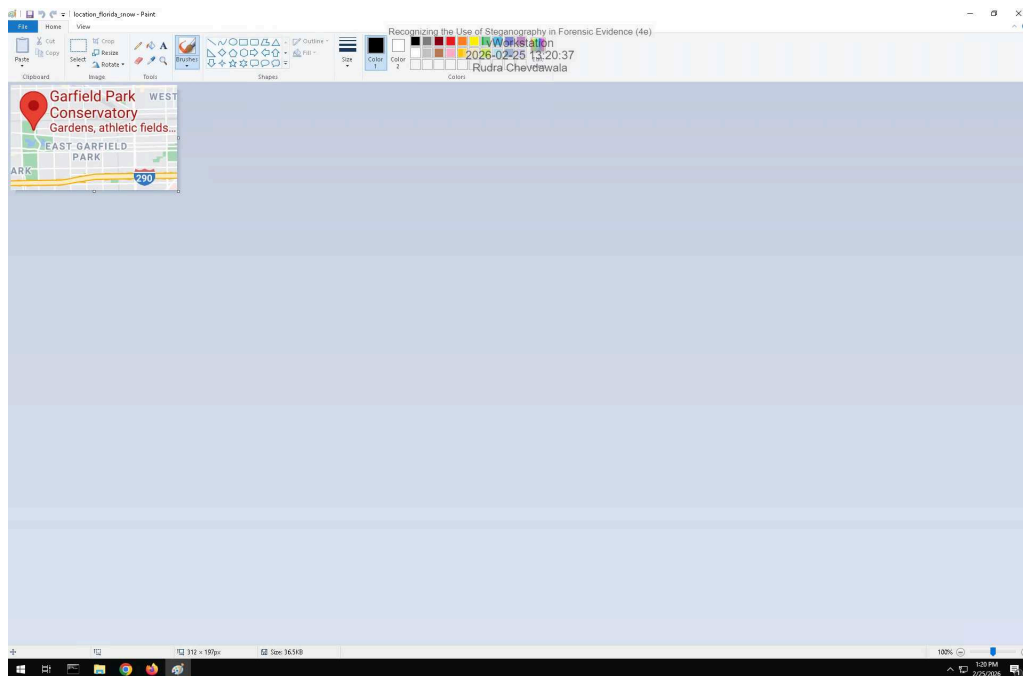
Record the names of the files that contain concealed data.

chicago.bmp is suspicious. Approximate amount of hidden data is 114062 bytes.

chicago1.bmp is suspicious. Approximate amount of hidden data is 57942 bytes.

Part 2: Extract More Hidden Data

Make a screen capture showing the **first file extracted by OpenStego**.



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Make a screen capture showing the second file extracted by OpenStego.

